

Binding Security of Combined KEMs

An Analysis of Real-World KEM Combiners

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Abstract. In this work we analyze the various binding properties of combiners for KEMs. We show that several binding properties follow easily for the most general combiner—assuming a collision-resistant hash function—while more performance-oriented combiners require the respective property from one or both of the underlying KEMs. Other binding properties are not obtained as directly and require either more properties of one underlying KEM or the respective property from both KEMs. Additionally, we provide an overview of the binding properties for several combinations of real-world KEMs.

1 Introduction

Key-encapsulation mechanisms (KEMs) generate a shared key k along with a ciphertext c , using a public key pk . With the corresponding secret key sk , the shared key k can be recovered from c , thereby enabling the receiver of a ciphertext to obtain the same shared key as the sender. The standard security notion for KEMs is IND-CCA—typically achieved via the Fujisaki-Okamoto transform [HHK17]—which ensures that c does not reveal information about k . Binding properties are additional security notions that turned out to be desired for hardening KEMs against attacks resulting from their usage in more complex protocols. Cremers et al. [CDM24] provided a framework, consisting of 6 *general notions* and 3 *attack models*, totalling 18 binding notions. For instance, the notion LEAK-BIND-K,PK-CT formalizes the idea that the shared key (K) and public key (PK) together bind the ciphertext (CT) in the leakage attack model (LEAK), meaning it should be infeasible for any adversary (that receives both public and secret key) to find two different ciphertexts that decapsulate to the same shared key.

Combiners allow to combine two or more KEMs into a new KEM. The idea is that the combined KEM should retain security as long as *at least one* of the underlying KEMs is secure. KEM combiners will play a crucial role in the

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transition towards post-quantum cryptography (PQC), when a classical KEM is combined with a post-quantum KEM. Such KEMs, which are referred to as hybrid KEMs, maintain security against classical attacks even if quantum computers break the classical KEM or the post-quantum KEM should turn out to be insecure.³ While KEM combiners are well-studied regarding the standard security notion IND-CCA [GHP18, BBF⁺19, HV21, CLM25, CP20], they have not been formally analyzed with respect to their binding properties.

1.1 Contribution

In this work, we close the aforementioned gap by analyzing the binding properties of KEM combiners. Our results are twofold as described in the following.

First, we consider a generic combiner—from which several real-world KEM combiners can be derived—and provide bounds regarding the 18 binding security notions. This combiner relies on the underlying KEMs to generate their shared keys and ciphertexts and then computes the final shared key by combining shared key, public key, and ciphertext of both KEMs via some generic function COMBINE. Specific KEM combiners can be obtained by specifying COMBINE and which of the inputs it uses: For instance, the OPENPGP combiner [KRSW25] is obtained by defining COMBINE to be SHA-256, applied to the shared keys, together with the ciphertext and public key from only one of the underlying KEMs. We show that the generic combiner can achieve all 12 binding notions of the form where the shared key binds other components, based on COMBINE being collision-resistant—if both the public keys and ciphertexts are inputs to COMBINE. If certain inputs to COMBINE are dropped, security is maintained if the respective KEM achieves the binding notion itself. For the remaining 6 notions, which formalize if the ciphertext binds the shared key or the public key, the results are qualitatively different. For the notions X-BIND-CT-PK and X-BIND-CT-K, security of the combiner is independent of COMBINE. In case of X-BIND-CT-PK, we obtain a “hybrid” bound, i.e., security of the combined KEM follows if at least one of the underlying KEMs is secure—though only in the HON and LEAK setting while in the MAL setting, both underlying KEMs need to be secure. For X-BIND-CT-K both KEMs need to be secure for the combined KEM to be X-BIND-CT-K, no matter the attack model. However, if one underlying KEM achieves X-BIND-CT-K *and* X-BIND-K,CT-PK, the combined KEM achieves X-BIND-CT-K security, though, again, only in the HON and LEAK setting.

Second, we apply our generic results to concrete KEM combiners and consider instantiations using KEMs for which the binding properties have been analyzed: ML-KEM, HQC, DH-KEM, FRODOKEM, BIKE, and CLASSIC-MCCELIECE. For these KEMs, the relevant binding notions are either achieved entirely or partially but never not at all. This implies that hashing all components (shared key, ciphertext, and public key) of one KEM is typically unnecessary to achieve all

³ Attacks against schemes in the NIST standardization process have shown that even schemes that have been scrutinized might exhibit vulnerabilities.

binding properties. Instead, it turns out that merely adding the public key is often sufficient, though none of the considered combiners is of that form.

1.2 Related Work

Anonymity, introduced in [BBDP01], ensures that a ciphertext does not reveal under which key it was generated. The notion of robust (public-key) encryption can be traced back to [ABN10, FLPQ13] and plays a crucial role in achieving anonymity. Binding security as formalized in [CDM24] generalizes the idea of robustness. Robustness analyses for the KEMs that were part of the NIST PQC standardization process [NIS17] are given in [GMP22, Xag22]. Binding properties of the implicitly-rejecting Fujisaki-Okamoto transform, together with an analysis of BIKE and HQC, are given in [KSW25].

Several works [GHP18, ADK⁺22, BBF⁺19, ABK25, CHH⁺25] describe and/or analyze various combiners for KEMs. Furthermore, there are numerous IETF drafts, e.g., [Jos25, CBG25, OGP⁺25, KRSW25, SFG25] which describe concrete KEM combiners.

Concurrent Work. Connolly et al. [CHH⁺25], in concurrent and independent work, generalize the X-WING combiner [BCD⁺24]. They give a detailed analysis for the generalized versions (QSF and QSI) both for standard IND-CCA security and binding security. The latter part overlaps with parts of our results with minimal difference in the bounds. In contrast to their work, we are focusing solely on binding security but consider a wider range of KEM combiners.

1.3 Organization

Section 2 provides the necessary background for this work. Our analysis of the generic combiner appears in Section 3, while the application of the results to specific KEM combiners is given in Section 4.

2 Preliminaries

Notation. We denote the empty string by ε . For a randomized algorithm A , we write $y \leftarrow^s A(x)$ to denote that y is the output of A on input x . We write $y \leftarrow A(x; r)$, to denote that y is the output of A on input x with randomness r —this means that A first generates some randomness r and then runs $A(x; r)$. For a deterministic algorithm, we simply write $y \leftarrow A(x)$. Throughout we leave the security parameter an implicit input for every algorithm. By *adversary*, we refer to a probabilistic algorithm running in polynomial-time (in the security parameter).

Key-Encapsulation Mechanisms. Below we provide the formal definition of key-encapsulation mechanisms.

Definition 1. A key-encapsulation mechanism is a tuple of three efficient algorithms (KEYGEN, ENCAPS, DECAPS). The (probabilistic) key-generation algorithm outputs a key pair (pk, sk) . The (probabilistic) encapsulation algorithm, on input a public key pk , outputs a ciphertext c and a shared key k . The (deterministic) decapsulation algorithm, on input a secret key sk ⁴ and a ciphertext c , outputs a shared key k . Correctness requires that $k = \bar{k}$ for $(c, k) \leftarrow \text{ENCAPS}(pk)$ and $\bar{k} \leftarrow \text{DECAPS}(sk, c)$ with $(pk, sk) \leftarrow \text{KEYGEN}()$.

Throughout this work, we will be concerned with two KEMs that are combined via some combiner. To differentiate the two, we will use subscripts 1 and 2. The subscripts are also used to differentiate the inputs/outputs of the different KEMs, e.g., KEM_i denotes a particular KEM while c_i denotes a ciphertext of KEM_i . Note that we will consider both classical and post-quantum KEMs as instantiations for KEM_1 and KEM_2 .

Binding Properties. In Fig. 1 we provide the generic binding security game from which the various concrete binding properties can be derived. Every notion is of the form X-BIND-P-Q. Here, $X \in \{\text{HON}, \text{LEAK}, \text{MAL}\}$ describes the attack model which describes how the keys are generated. Simply speaking, HON and LEAK use honestly generated keys, the difference being that HON provides only the public key to the adversary while LEAK provides also the secret key. In case of $X = \text{MAL}$, the adversary can generate the keys itself and is thus no longer limited to honestly generated ones. The components P and Q describe which values (P) “bind” which other values (Q). For instance, X-BIND-K-CT captures the idea that the shared key (K) binds the ciphertext (CT) which, roughly speaking, formalizes the idea that it is hard to find two different ciphertexts that decapsulate to the same shared key. Similarly, X-BIND-K,CT-PK formalizes that it is hard to find a ciphertexts that decapsulates to the same shared key under different key pairs. We refer to [KSW25] for explicit descriptions of the concrete binding properties. Below we define security for the various binding properties.

Definition 2. Let KEM be a KEM and X-BIND-P-Q the security game shown in Fig. 1, where $X \in \{\text{HON}, \text{LEAK}, \text{MAL}\}$, $P \in \{\{K\}, \{CT\}, \{K, CT\}, \{K, PK\}\}$, and $Q \in \{\{K\}, \{PK\}, \{CT\}\}$. We say that KEM achieves X-BIND-P-Q if for any adversary $\mathcal{A}$, its probability in winning X-BIND-P-Q is negligible.

The following result from [CDM24] establishes a hierarchy between two binding notions that we will use later.

Lemma 3. Let KEM be a KEM and $X \in \{\text{LEAK}, \text{HON}\}$. Then for any adversary $\mathcal{A}$, there exists an adversary $\mathcal{B}$ such that

$$\text{Adv}_{\text{KEM}}^{\text{X-BIND-CT-K}}(\mathcal{A}) \leq \text{Adv}_{\text{KEM}}^{\text{X-BIND-CT-PK}}(\mathcal{B}).$$

⁴ Note that we assume that the public key can be derived from the corresponding secret key for honestly generated key pairs.

Game X-BIND-P-Q	Game MAL-BIND-P-Q
$(pk, sk) \leftarrow \text{KEYGEN}()$	$g \leftarrow \mathcal{A}()$
$(\overline{pk}, \overline{sk}) \leftarrow \text{KEYGEN}()$	if $g = 1$: // Scenario (I): DECAPS-DECAPS
if $PK \in P$:	$(sk, \overline{sk}, c, \overline{c}) \leftarrow \mathcal{A}()$
$(\overline{pk}, \overline{sk}) \leftarrow (pk, sk)$	$k \leftarrow \text{DECAPS}(sk, c)$
if $PK \notin P \cup Q$:	$\overline{k} \leftarrow \text{DECAPS}(\overline{sk}, \overline{c})$
$b \in \{0, 1\} \leftarrow \mathcal{A}()$	if $g = 2$: // Scenario (II): ENCAPS-DECAPS
if $b = 0$: $(\overline{pk}, \overline{sk}) \leftarrow (pk, sk)$	$(pk, \overline{sk}, r, \overline{c}) \leftarrow \mathcal{A}()$
if $X = \text{HON}$:	$(k, c) \leftarrow \text{ENCAPS}(pk; r)$
$(c, \overline{c}) \leftarrow \mathcal{A}^{\text{Dec}, \overline{\text{Dec}}}(pk, \overline{pk})$	$\overline{k} \leftarrow \text{DECAPS}(\overline{sk}, \overline{c})$
if $X = \text{LEAK}$:	if $g \notin \{1, 2\}$: // Scenario (III): ENCAPS-ENCAPS
$(c, \overline{c}) \leftarrow \mathcal{A}(pk, sk, \overline{pk}, \overline{sk})$	$(pk, \overline{pk}, r, \overline{r}) \leftarrow \mathcal{A}()$
$k \leftarrow \text{DECAPS}(sk, c)$	$(k, c) \leftarrow \text{ENCAPS}(pk; r)$
$\overline{k} \leftarrow \text{DECAPS}(\overline{sk}, \overline{c})$	$(\overline{k}, \overline{c}) \leftarrow \text{ENCAPS}(\overline{pk}; \overline{r})$
if $k = \perp \vee \overline{k} = \perp$:	if $k = \perp \vee \overline{k} = \perp$:
return 0	return 0
$v_p \leftarrow (\forall p \in P : x_p = \overline{x}_p)$	$v_p \leftarrow (\forall p \in P : x_p = \overline{x}_p)$
$v_q \leftarrow (\exists q \in Q : x_q \neq \overline{x}_q)$	$v_q \leftarrow (\exists q \in Q : x_q \neq \overline{x}_q)$
return $v_p \wedge v_q$	return $v_p \wedge v_q$

Fig. 1. Generic security games for the notions X-BIND-P-Q and MAL-BIND-P-Q for $P = \{K, \text{CT}, \{K, \text{CT}\}, \{K, \text{PK}\}\}$, $Q = \{K, \text{PK}, \text{CT}\}$, and $X \in \{\text{HON}, \text{LEAK}\}$. Note that for $p \in P$ (and $q \in Q$), we denote the corresponding instances by x_p , $\overline{x}_p$ and x_q , $\overline{x}_q$, respectively. For example, if $p = \text{CT}$, we have $x_p = c$ and $\overline{x}_p = \overline{c}$. For variants (I) and (II) of the MAL notions, it is understood that any check of the form $pk \bowtie \overline{pk}$, for $\bowtie \in \{=, \neq\}$ is performed on the public keys contained in the respective secret keys sk and $\overline{sk}$. The decapsulation oracles Dec and $\overline{\text{Dec}}$ in case of $X = \text{HON}$ are omitted and work in the obvious way (without any restrictions) using sk and $\overline{sk}$, respectively.

3 Binding Analysis of Combined KEMs

In this section, we present our results regarding the binding notions of combined KEMs. For this, we introduce a generic combiner in [Section 3.1](#), which covers all real-world KEM combiners we are aware of. In [Section 3.2](#), we analyze the generic combiner with respect to binding notions X-BIND-K-CT, X-BIND-K-PK, X-BIND-K,PK-CT, and X-BIND-K,CT-PK, i.e., the cases in which the shared key is binding. The remaining notions X-BIND-CT-K and X-BIND-CT-PK, i.e., the ones where the ciphertext by itself is binding, are treated in [Section 3.3](#). Note that this covers all notions presented in [\[CDM24\]](#).

3.1 A Generic Combiner

We define a generic combiner GC, which combines two KEMs (KEM_1 and KEM_2) into a combined KEM cKEM as displayed in Fig. 2. For key generation, the combiner independently calls the key generation algorithms of the underlying KEMs KEM_1 and KEM_2 , and then assembles the public and secret key of the combined scheme as $pk = (pk_1, pk_2)$ and $sk = (sk_1, sk_2)$. During encapsulation, the ENCAPS algorithms of KEM_1 and KEM_2 are used to generate the ciphertexts c_1 , c_{pk} and shared keys k_1 , k_2 of the underlying KEMs. The ciphertext of the combined scheme is obtained simply as $c = (c_1, c_2)$ —in contrast, the computation of the shared key differs depending on the concrete combiner. While the shared keys are always used in the key derivation, this is not necessarily the case for the public keys and ciphertexts of the underlying schemes. To model this, we introduce an abstract function COMBINE that is used to generate the shared key and a filtering function F that describes which of the components (pk_1, pk_2, c_1, c_2) are given as input to COMBINE . Thus, a key encapsulation mechanism cKEM built via the generic combiner GC, computes the shared secret as

$$\text{COMBINE}(k_1, k_2, F(pk_1, pk_2, c_1, c_2), \Xi)$$

where Ξ describes an abstract label⁵ and F filters out the components that are not needed for the combining operation of cKEM . More precisely, F takes as input the tuple (pk_1, pk_2, c_1, c_2) and outputs a sub-tuple. As an example, for the combiner underlying X-WING, we have $\text{COMBINE} = \text{SHA-256}$ and $F(pk_1, pk_2, c_1, c_2) = (pk_1, c_1)$. Table 1 provides an overview of several combiners which can be derived as specific instantiations of our generic combiner.

For ease of exposition, we also consider a second filtering function to track the inputs that are included in the key derivation. For cKEM , the function

$$f: \{\text{PK}, \text{CT}\} \times \{1, 2\} \rightarrow \{0, 1\}$$

outputs 1 if and only if the component corresponding to the input is part of key derivation. For instance, referring to the X-WING example above, where $F(pk_1, pk_2, c_1, c_2) = (pk_1, c_1)$, we have $f(\text{PK}, 1) = f(\text{CT}, 1) = 1$ and $f(\text{PK}, 2) = f(\text{CT}, 2) = 0$.

A key-encapsulation mechanism cKEM constructed from the generic combiner GC with filtering function f is denoted by $\text{cKEM}[f]$.

Generic Reductions. For the results in the following, we rely on some generic reductions, reducing a certain binding property of a combined KEM to a binding property of one of the underlying KEMs. We identify these reductions by $\mathcal{H}_i$, $\mathcal{L}_i$, and $\mathcal{M}_i$ —the letter indicating for which attack model (HON, LEAK, MAL) the reduction is. These reductions work in a straightforward way. For instance, $\mathcal{L}_1$ is a reduction playing against some binding notion (in the LEAK setting) of KEM_1 . In case of LEAK-BIND-K-CT, it receives two key pairs (pk_1, sk_1) , (pk_1, sk_1) and has

⁵ Note that for our analysis Ξ is not relevant.

KEYGEN()	ENCAPS($pk = (pk_1, pk_2)$)
$(pk_1, sk_1) \leftarrow \$ \text{KEYGEN}_1()$	$(c_1, k_1) \leftarrow \$ \text{ENCAPS}_1(pk_1)$
$(pk_2, sk_2) \leftarrow \$ \text{KEYGEN}_2()$	$(c_2, k_2) \leftarrow \$ \text{ENCAPS}_2(pk_2)$
$pk \leftarrow (pk_1, pk_2)$	$c \leftarrow (c_1, c_2)$
$sk \leftarrow (sk_1, sk_2)$	$k \leftarrow \text{COMBINE}(k_1, k_2, F(pk_1, pk_2, c_1, c_2), \Xi)$
return (pk, sk)	return (c, k)
	DECAPS($sk = (sk_1, sk_2), c = (c_1, c_2)$)
	$k_1 \leftarrow \text{DECAPS}_1(sk_1, c_1)$
	$k_2 \leftarrow \text{DECAPS}_2(sk_2, c_2)$
	$k \leftarrow \text{COMBINE}(k_1, k_2, F(pk_1, pk_2, c_1, c_2), \Xi)$
	return k

Fig. 2. Generic combiner GC which combines two KEMs KEM_1 and KEM_2 into a KEM $\text{cKEM}[f]$, where f describes which inputs are filtered out by F .

to output two ciphertexts $c_1 \neq \bar{c}_1$. This reduction runs an adversary $\mathcal{A}$ playing against LEAK-BIND-K-CT security of the *combined* KEM cKEM , by providing its own key pairs alongside two key pairs of KEM_2 which $\mathcal{L}_1$ locally generates. When $\mathcal{A}$ outputs two distinct ciphertexts $c = (c_1, c_2)$ and $\bar{c} = (\bar{c}_1, \bar{c}_2)$, $\mathcal{L}_1$ outputs the two ciphertexts c_1 and $\bar{c}_1$. For notions that let the adversary decide if it wants to play against two distinct key pairs or a single one (e.g., LEAK-BIND-CT-K), the reduction will generate one or two key pairs locally depending on the choice of $\mathcal{A}$ and also ask for the same number of key pairs in its own security game. Simulation of the decapsulation oracles in the HON setting is straightforward as $\mathcal{H}_i$ can use its own decapsulation oracle for everything related to KEM_i and can locally simulate decapsulation for the other KEM as it has access to the secret key.

3.2 Results for Shared Key Binding

In this section, we present our result regarding the X-BIND-K-Q security of the generic combiner. Simply put, a combined KEM achieves X-BIND-K-Q, if COMBINE is collision-resistant (we denote by CR the collision-resistance game), and either COMBINE includes the Q-component of the underlying KEM KEM_i or KEM_i is itself X-BIND-K-Q secure.

Theorem 4. *Let KEM_1 and KEM_2 be two KEMs. Let further cKEM denote the KEM resulting from applying GC to KEM_1 and KEM_2 (with combiner function COMBINE and filtering function f). Furthermore, let $X \in \{\text{HON}, \text{MAL}, \text{LEAK}\}$, $P \in \{\{K\}, \{K, \text{CT}\}, \{K, \text{PK}\}\}$, and $Q \in \{\{\text{CT}\}, \{\text{PK}\}\}$, such that $P \cap Q = \emptyset$.*

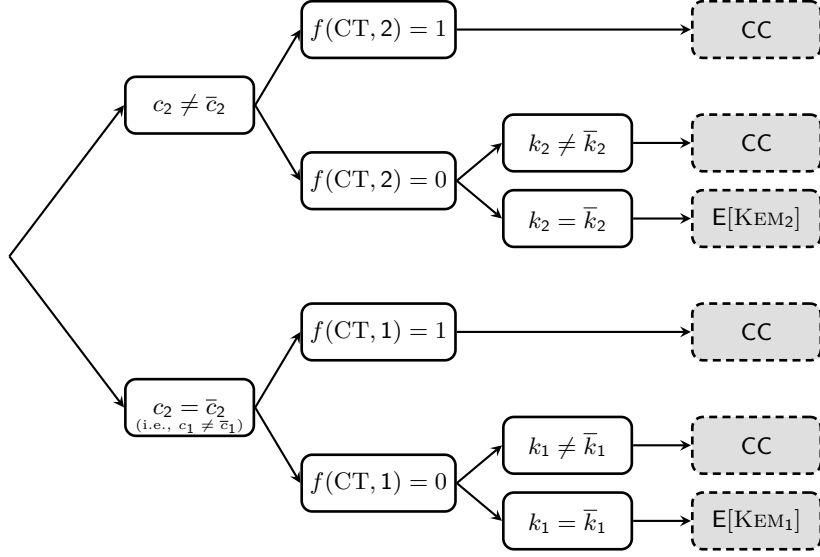


Fig. 3. Case distinction for the proof of [Theorem 4](#). The gray boxes show which of the events (collision of COMBINE (CC) or X-BIND-P-CT security of KEM_i being broken ($\text{E}[\text{KEM}_i]$)) occurs in each case.

Then for any adversary $\mathcal{A}$ there exist adversaries $\mathcal{B}$, $\mathcal{C}_1$, and $\mathcal{C}_2$, such that

$$\text{Adv}_{\text{cKEM}[f]}^{\text{X-BIND-P-Q}}(\mathcal{A}) \leq 2 \text{Adv}_{\text{COMBINE}}^{\text{CR}}(\mathcal{B}) + \sum_{\substack{i \in \{1,2\} \\ f(Q,i)=0}} \text{Adv}_{\text{KEM}_i}^{\text{X-BIND-P-Q}}(\mathcal{C}_i).$$

Proof. Let $\mathcal{A}$ be an adversary that breaks X-BIND-P-Q security of $\text{cKEM}[f]$. Since $\text{K} \in \text{P}$, the outputs of the adversary $\mathcal{A}$ against X-BIND-P-Q security of $\text{cKEM}[f]$ yield colliding keys, i.e.,

$$\begin{aligned} k &= \text{COMBINE}(k_1, k_2, F(pk_1, pk_2, c_1, c_2), \Xi) \\ &= \text{COMBINE}(\bar{k}_1, \bar{k}_2, F(\bar{p}k_1, \bar{p}k_2, \bar{c}_1, \bar{c}_2), \Xi) = \bar{k}. \end{aligned}$$

Further, depending on the value of $Q \in \{\{\text{CT}\}, \{\text{PK}\}\}$, either $(c_1, c_2) \neq (\bar{c}_1, \bar{c}_2)$ or $(pk_1, pk_2) \neq (\bar{p}k_1, \bar{p}k_2)$ must hold. For sake of simplicity, we focus on the case that $Q = \{\text{CT}\}$ and note that the argumentation for $Q = \{\text{PK}\}$ works analogously.

Thus, we know that $(c_1, c_2) \neq (\bar{c}_1, \bar{c}_2)$. For $i \in \{1, 2\}$, we distinguish the cases shown in [Fig. 3](#). Here, CC denotes a collision of COMBINE—note that if collision-resistance (CR) is given, the probability for CC is negligible—and the event $\text{E}[\text{KEM}_i]$ describes that X-BIND-P-CT security of KEM_i has been broken.

Since we know that $k = \bar{k}$, a collision of COMBINE appears if any of the inputs differ—taking into account which inputs are filtered out by f . In the remaining two cases in [Fig. 3](#), the events $\text{E}[\text{KEM}_1]$ and $\text{E}[\text{KEM}_2]$ appear. Note that in the

case that $P = \{K, PK\}$, the fact that $\mathcal{A}$ breaks the notion X-BIND-P-Q for $\text{cKEM}[f]$, implies that the public keys of the underlying KEMs also agree as $pk = (pk_1, pk_2)$. In total, taking into account all cases from the above tree, yields the claimed bound as follows.

Firstly, mirroring the above tree, we divide the advantage based on whether $c_2 = \bar{c}_2$ or $c_2 \neq \bar{c}_2$ holds. In the following let **Win** denote the event that $\mathcal{A}$ wins the game X-BIND-P-CT:

$$\mathbf{Adv}_{\text{cKEM}[f]}^{\text{X-BIND-P-CT}}(\mathcal{A}) = \Pr[\mathbf{Win}] = \Pr[\mathbf{Win} \wedge c_2 \neq \bar{c}_2] + \Pr[\mathbf{Win} \wedge c_2 = \bar{c}_2]$$

Next, we further split up both summands, based on the values of $f(\text{CT}, 2)$ and $f(\text{CT}, 1)$ —corresponding to the second branching point in the tree:

$$\begin{aligned} \Pr[\mathbf{Win} \wedge c_2 \neq \bar{c}_2] &= \Pr[\mathbf{Win} \wedge c_2 \neq \bar{c}_2 \wedge f(\text{CT}, 2) = 1] \\ &\quad + \Pr[\mathbf{Win} \wedge c_2 \neq \bar{c}_2 \wedge f(\text{CT}, 2) = 0] \\ \Pr[\mathbf{Win} \wedge c_2 = \bar{c}_2] &= \Pr[\mathbf{Win} \wedge c_2 = \bar{c}_2 \wedge f(\text{CT}, 1) = 1] \\ &\quad + \Pr[\mathbf{Win} \wedge c_2 = \bar{c}_2 \wedge f(\text{CT}, 1) = 0] \end{aligned}$$

Lastly, the two summands with $f(\text{CT}, 2) = 0$ and $f(\text{CT}, 1) = 0$ are further divided according to the third branching point in the above tree:

$$\begin{aligned} &\Pr[\mathbf{Win} \wedge c_2 \neq \bar{c}_2 \wedge f(\text{CT}, 2) = 0] \\ &= \Pr[\mathbf{Win} \wedge c_2 \neq \bar{c}_2 \wedge f(\text{CT}, 2) = 0 \wedge k_2 \neq \bar{k}_2] \\ &\quad + \Pr[\mathbf{Win} \wedge c_2 \neq \bar{c}_2 \wedge f(\text{CT}, 2) = 0 \wedge k_2 = \bar{k}_2] \\ &\Pr[\mathbf{Win} \wedge c_1 \neq \bar{c}_1 \wedge f(\text{CT}, 1) = 0] \\ &= \Pr[\mathbf{Win} \wedge c_1 \neq \bar{c}_1 \wedge f(\text{CT}, 1) = 0 \wedge k_1 \neq \bar{k}_1] \\ &\quad + \Pr[\mathbf{Win} \wedge c_1 \neq \bar{c}_1 \wedge f(\text{CT}, 1) = 0 \wedge k_1 = \bar{k}_1] \end{aligned}$$

Combining these results and using the formula for conditional probability yields

$$\begin{aligned} &\mathbf{Adv}_{\text{cKEM}[f]}^{\text{X-BIND-P-CT}}(\mathcal{A}) \\ &= \Pr[\mathbf{Win} \wedge c_2 \neq \bar{c}_2 \mid f(\text{CT}, 2) = 1] \cdot \Pr[f(\text{CT}, 2) = 1] \\ &\quad + \Pr[\mathbf{Win} \wedge c_2 \neq \bar{c}_2 \wedge k_2 \neq \bar{k}_2 \mid f(\text{CT}, 2) = 0] \cdot \Pr[f(\text{CT}, 2) = 0] \\ &\quad + \Pr[\mathbf{Win} \wedge c_2 \neq \bar{c}_2 \wedge k_2 = \bar{k}_2 \mid f(\text{CT}, 2) = 0] \cdot \Pr[f(\text{CT}, 2) = 0] \\ &\quad + \Pr[\mathbf{Win} \wedge c_2 = \bar{c}_2 \mid f(\text{CT}, 1) = 1] \cdot \Pr[f(\text{CT}, 1) = 1] \\ &\quad + \Pr[\mathbf{Win} \wedge c_1 \neq \bar{c}_1 \wedge k_1 \neq \bar{k}_1 \mid f(\text{CT}, 1) = 0] \cdot \Pr[f(\text{CT}, 1) = 0] \\ &\quad + \Pr[\mathbf{Win} \wedge c_1 \neq \bar{c}_1 \wedge k_1 = \bar{k}_1 \mid f(\text{CT}, 1) = 0] \cdot \Pr[f(\text{CT}, 1) = 0]. \end{aligned}$$

Note that for any given combiner, the values of $f(\text{CT}, 2)$ and $f(\text{CT}, 1)$ are pre-determined, i.e., $\Pr[f(\text{CT}, 2) = 0], \Pr[f(\text{CT}, 1) = 0] \in \{0, 1\}$. Using this and the

events CC , $\text{E}[\text{KEM}_2]$, and $\text{E}[\text{KEM}_1]$ yields for $\mathcal{C}_i \in \{\mathcal{H}_i, \mathcal{L}_i, \mathcal{M}_i\}$ the final bound

$$\begin{aligned}
\text{Adv}_{\text{CKEM}[f]}^{\text{X-BIND-P-CT}}(\mathcal{A}) &\leq \Pr[\text{CC}] \cdot \Pr[f(\text{CT}, 2) = 1] \\
&\quad + \Pr[\text{CC}] \cdot \Pr[f(\text{CT}, 2) = 0] \\
&\quad + \Pr[\text{E}[\text{KEM}_2]] \cdot \Pr[f(\text{CT}, 2) = 0] \\
&\quad + \Pr[\text{CC}] \cdot \Pr[f(\text{CT}, 1) = 1] \\
&\quad + \Pr[\text{CC}] \cdot \Pr[f(\text{CT}, 1) = 0] \\
&\quad + \Pr[\text{E}[\text{KEM}_1]] \cdot \Pr[f(\text{CT}, 1) = 0] \\
&= 2 \text{Adv}_{\text{COMBINE}}^{\text{CR}}(\mathcal{B}) + \sum_{\substack{i \in \{1,2\} \\ f(\text{CT}, i) = 0}} \text{Adv}_{\text{KEM}_i}^{\text{X-BIND-P-CT}}(\mathcal{C}_i).
\end{aligned}$$

As this holds analogously for $\mathcal{Q} = \{\text{PK}\}$, we obtain the theorem statement

$$\text{Adv}_{\text{CKEM}[f]}^{\text{X-BIND-P-Q}}(\mathcal{A}) \leq 2 \text{Adv}_{\text{COMBINE}}^{\text{CR}}(\mathcal{B}) + \sum_{\substack{i \in \{1,2\} \\ f(\mathcal{Q}, i) = 0}} \text{Adv}_{\text{KEM}_i}^{\text{X-BIND-P-Q}}(\mathcal{C}_i).$$

□

To illustrate the above theorem, we give an exemplary application for the notion HON-BIND-K-CT and the OPENPGP combiner [KRSW25], for which $\text{COMBINE} = \text{SHA-256}$ and $F(pk_1, pk_2, c_1, c_2) = (pk_1, c_1)$ holds. This yields the bound

$$\text{Adv}_{\text{CKEM}}^{\text{HON-BIND-K-CT}}(\mathcal{A}) \leq 2 \text{Adv}_{\text{SHA-256}}^{\text{CR}}(\mathcal{B}) + \text{Adv}_{\text{KEM}_2}^{\text{HON-BIND-K-CT}}(\mathcal{C}),$$

which implies that the KEM CKEM obtained from the OPENPGP combiner is HON-BIND-K-CT secure if KEM_2 is, and collision-resistance of SHA-256 is given.

3.3 Results for Ciphertext Binding

In this section, we give our results regarding the X-BIND-CT-Q security of the generic combiner. In Theorem 5, we treat X-BIND-CT-PK , while Corollary 6 and Theorem 7 concern X-BIND-CT-K . Note that these notions are unachievable for any KEM using implicit rejection as shown in [CDM24]. Looking ahead, the KEMs we will analyze in Section 4 all use implicit rejection, making the results of this section inapplicable. Nevertheless, they are of relevance for any combination of explicitly-rejecting KEMs.

The theorem below shows under which conditions the ciphertext binds the public key. In the HON and LEAK setting, having the property by at least one of the underlying KEMs is sufficient for the combined KEM to inherit it. In the MAL setting, the property is required from both underlying KEMs as the adversary can freely choose the key pairs: if the property is easy to break for one KEM, the adversary can simply choose distinct key pairs which only disagree on the easy-to-break KEM.

Theorem 5. Let KEM_2 and KEM_1 be two KEMs. Let further cKEM denote the KEM resulting from applying GC to KEM_2 and KEM_1 . Then for any adversary $\mathcal{A}$ there exist adversaries $\mathcal{M}_i$, $\mathcal{L}_i$, and $\mathcal{H}_i$ for $i \in \{1, 2\}$, such that

$$\text{Adv}_{\text{cKEM}}^{\text{X-BIND-CT-PK}}(\mathcal{A}) \leq \begin{cases} \sum_{i \in \{1, 2\}} \text{Adv}_{\text{KEM}_i}^{\text{MAL-BIND-CT-PK}}(\mathcal{M}_i) & \text{if } X = \text{MAL}, \\ \min_{i \in \{1, 2\}} \text{Adv}_{\text{KEM}_i}^{\text{LEAK-BIND-CT-PK}}(\mathcal{L}_i) & \text{if } X = \text{LEAK}, \\ \min_{i \in \{1, 2\}} \text{Adv}_{\text{KEM}_i}^{\text{HON-BIND-CT-PK}}(\mathcal{H}_i) & \text{if } X = \text{HON}. \end{cases}$$

Proof. Let $\mathcal{A}$ be an adversary that breaks X-BIND-CT-PK security of cKEM . We distinguish three cases based on the attack model:

Case 1: $X = \text{MAL}$

Adversary $\mathcal{A}$ outputs two key pairs $(pk, sk), (\overline{pk}, \overline{sk})$, where $pk = (pk_1, pk_2)$, $sk = (sk_1, sk_2)$, $\overline{pk} = (\overline{pk}_1, \overline{pk}_2)$, and $\overline{sk} = (\overline{sk}_1, \overline{sk}_2)$. For $\mathcal{A}$ to be successful w.r.t. X-BIND-CT-PK, it is required that $pk \neq \overline{pk}$, which implies that *at least one* of the following two inequalities has to hold: $pk_1 \neq \overline{pk}_1$ or $pk_2 \neq \overline{pk}_2$. Thus we get

$$\begin{aligned} \text{Adv}_{\text{cKEM}}^{\text{X-BIND-CT-PK}}(\mathcal{A}) &\leq \Pr[\mathcal{A} \text{ wins X-BIND-CT-PK} \wedge pk_1 \neq \overline{pk}_1] \\ &\quad + \Pr[\mathcal{A} \text{ wins X-BIND-CT-PK} \wedge pk_2 \neq \overline{pk}_2]. \end{aligned}$$

In the following, we bound the first term; bounding the second term works completely analogue. For a successful $\mathcal{A}$ there are three scenarios: DECAPS–DECAPS, ENCAPS–DECAPS, and ENCAPS–ENCAPS. We give the details for the first scenario (DECAPS–DECAPS) and subsequently discuss how it extends to the others. In the DECAPS–DECAPS scenario, $\mathcal{A}$ outputs a single ciphertext c which decapsulates under both secret keys to some valid shared keys, i.e.,

$$\begin{aligned} \perp &\neq k \leftarrow \text{DECAPS}(sk, c), \text{ and} \\ \perp &\neq \overline{k} \leftarrow \text{DECAPS}(\overline{sk}, c). \end{aligned}$$

By construction, this means that the ciphertexts of the underlying KEMs validly decapsulate under the underlying secret keys, i.e., for $i \in \{1, 2\}$, we have

$$\perp \neq k_i \leftarrow \text{DECAPS}(sk_i, c_i).$$

Using $\mathcal{M}_1$ from our generic reductions then yields

$$\Pr[\mathcal{A} \text{ wins X-BIND-CT-PK} \wedge pk_1 \neq \overline{pk}_1] \leq \text{Adv}_{\text{KEM}_1}^{\text{X-BIND-CT-PK}}(\mathcal{M}_1).$$

In the other scenarios, $\mathcal{A}$ either outputs a ciphertext and a secret key for the first tuple and a public key and random coins for the second tuple (ENCAPS–DECAPS) or two tuples consisting of public key and random coins each (ENCAPS–ENCAPS). Regardless of which, the ciphertexts are required to be equal and must decrypt to valid shared keys. Our adversary $\mathcal{M}_1$ can simply take the parts belonging to KEM_1 from $\mathcal{A}$'s output to also break MAL-BIND-CT-PK of KEM_1 (in the

same scenario as $\mathcal{A}$ breaks MAL-BIND-CT-PK of cKEM). For the second term, a similar argument yields

$$\Pr[\mathcal{A} \text{ wins X-BIND-CT-PK} \wedge pk_2 \neq \overline{pk_2}] \leq \mathbf{Adv}_{\text{KEM}_2}^{\text{X-BIND-CT-PK}}(\mathcal{M}_2).$$

Collecting the above then yields

$$\begin{aligned} \mathbf{Adv}_{\text{cKEM}}^{\text{X-BIND-CT-PK}}(\mathcal{A}) &\leq \sum_{i \in \{1,2\}} \Pr[\mathcal{A} \text{ wins X-BIND-CT-PK} \wedge pk_i \neq \overline{pk_i}] \\ &\leq \sum_{i \in \{1,2\}} \mathbf{Adv}_{\text{KEM}_i}^{\text{X-BIND-CT-PK}}(\mathcal{M}_i). \end{aligned}$$

Case 2: X = LEAK

Adversary $\mathcal{A}$ receives two key pairs $(pk, sk), (\overline{pk}, \overline{sk})$, where $pk = (pk_1, pk_2)$, $sk = (sk_1, sk_2)$, $\overline{pk} = (\overline{pk}_1, \overline{pk}_2)$, and $\overline{sk} = (\overline{sk}_1, \overline{sk}_2)$. With overwhelming probability, we have $pk_i \neq \overline{pk}_i$ and $sk_i \neq \overline{sk}_i$ for $i \in \{1, 2\}$. Adversary $\mathcal{A}$ outputs a *single* ciphertext $c = (c_1, c_2)$ such that

$$\begin{aligned} \perp &\neq k \leftarrow \text{cKEM}(sk, c) \text{ and} \\ \perp &\neq \overline{k} \leftarrow \text{cKEM}(\overline{sk}, c). \end{aligned}$$

By construction, this implies, for $i \in \{1, 2\}$, that

$$\begin{aligned} \perp &\neq k_i \leftarrow \text{KEM}_i(sk_i, c_i) \text{ and} \\ \perp &\neq \overline{k}_i \leftarrow \text{KEM}_i(\overline{sk}_i, c_i), \end{aligned}$$

as otherwise one of the inputs to COMBINE would be $\perp$. For $\mathcal{L}_i$ from our generic reductions, we then obtain

$$\begin{aligned} \mathbf{Adv}_{\text{cKEM}}^{\text{LEAK-BIND-CT-PK}}(\mathcal{A}) &\leq \mathbf{Adv}_{\text{KEM}_1}^{\text{LEAK-BIND-CT-PK}}(\mathcal{L}_1) \text{ and} \\ \mathbf{Adv}_{\text{cKEM}}^{\text{LEAK-BIND-CT-PK}}(\mathcal{A}) &\leq \mathbf{Adv}_{\text{KEM}_2}^{\text{LEAK-BIND-CT-PK}}(\mathcal{L}_2), \end{aligned}$$

which yields

$$\mathbf{Adv}_{\text{cKEM}}^{\text{LEAK-BIND-CT-PK}}(\mathcal{A}) \leq \min_{i \in \{1,2\}} \mathbf{Adv}_{\text{KEM}_i}^{\text{LEAK-BIND-CT-PK}}(\mathcal{L}_i).$$

Case 3: X = HON

The HON setting works essentially like the LEAK setting. The mere difference is that $\mathcal{A}$ has access to a decapsulation oracle which the reduction can simulate by locally evaluating every computation related to the KEM that it generated itself and its own decapsulation oracle for every computation related to the KEM that it plays against. $\square$

The hierarchy between X-BIND-CT-PK and X-BIND-CT-K (cf. [Lemma 3](#)) in the HON and LEAK setting allows us to derive a bound for HON-BIND-CT-K and LEAK-BIND-CT-K from [Theorem 5](#).

Corollary 6. *Let KEM_1 and KEM_2 be two KEMs. Let further cKEM denote the KEM resulting from applying GC to KEM_1 and KEM_2 . Then for any adversary $\mathcal{A}$ and $X \in \{\text{HON}, \text{LEAK}\}$, there exist adversaries $\mathcal{B}_1$ and $\mathcal{B}_2$, such that*

$$\text{Adv}_{\text{cKEM}}^{\text{X-BIND-CT-K}}(\mathcal{A}) \leq \min_{i \in \{1,2\}} \text{Adv}_{\text{KEM}_i}^{\text{X-BIND-CT-PK}}(\mathcal{B}_i).$$

Proof. This follows directly from Lemma 3 and Theorem 5. $\square$

Although the above corollary yields an initial bound for HON-BIND-CT-K and LEAK-BIND-CT-K, a better bound—using weaker assumptions on the underlying KEMs—can be derived through a dedicated analysis. Further, the results so far do not cover MAL-BIND-CT-K.

The theorem below provides two bounds on the X-BIND-CT-K security of a KEM obtained via the generic combiner GC, subject to the public keys *not* being an input to the combine function COMBINE. The first bound holds for $X \in \{\text{HON}, \text{LEAK}\}$ and states that if *any* of the two underlying KEMs is both X-BIND-CT-K and X-BIND-K,CT-PK secure, then the combined KEM is X-BIND-CT-K secure. The second bound covers $X \in \{\text{HON}, \text{LEAK}, \text{MAL}\}$ and states that the combined KEM is X-BIND-CT-K secure if *both* of the underlying KEMs are X-BIND-CT-K secure.

Theorem 7. *Let KEM_1 and KEM_2 be two KEMs. Let further cKEM denote the KEM resulting from applying GC to KEM_1 and KEM_2 (with combiner function COMBINE and input-selection function f for which $f(\text{PK}, \cdot) = 0$). Then for any adversary $\mathcal{A}$, there exist adversaries $\mathcal{B}_i$ and $\mathcal{C}_i$ for $i \in \{1,2\}$, such that the following bounds hold*

(1) for $X \in \{\text{HON}, \text{LEAK}\}$:

$$\begin{aligned} & \text{Adv}_{\text{cKEM}[f]}^{\text{X-BIND-CT-K}}(\mathcal{A}) \\ & \leq \min_{i \in \{1,2\}} \left(2 \text{Adv}_{\text{KEM}_i}^{\text{X-BIND-CT-K}}(\mathcal{B}_i) + \text{Adv}_{\text{KEM}_i}^{\text{X-BIND-K,CT-PK}}(\mathcal{C}_i) \right) \end{aligned}$$

(2) for $X \in \{\text{HON}, \text{LEAK}, \text{MAL}\}$:

$$\text{Adv}_{\text{cKEM}[f]}^{\text{X-BIND-CT-K}}(\mathcal{A}) \leq 2 \text{Adv}_{\text{KEM}_1}^{\text{X-BIND-CT-K}}(\mathcal{B}_1) + 2 \text{Adv}_{\text{KEM}_2}^{\text{X-BIND-CT-K}}(\mathcal{B}_2).$$

Proof. We first handle the HON and LEAK case and subsequently discuss how the results extend to the MAL setting. Let $\mathcal{A}$ be an adversary against the X-BIND-CT-K security of cKEM for $X \in \{\text{HON}, \text{LEAK}\}$. This adversary receives two honestly generated public keys $pk, \overline{pk}$ (in the case that $X = \text{HON}$) and two key pairs $(pk, sk), (\overline{pk}, \overline{sk})$ (in the case that $X = \text{LEAK}$), where $pk = (pk_1, pk_2)$, $sk = (sk_1, sk_2)$, $\overline{pk} = (\overline{pk}_1, \overline{pk}_2)$, and $\overline{sk} = (\overline{sk}_1, \overline{sk}_2)$.⁶ It outputs a ciphertext

⁶ The X-BIND-CT-K notion technically allows $\mathcal{A}$ to also play against just a single key pair by setting the key pairs to be the same. In this case, however, $\mathcal{A}$ cannot win because it requires that the (deterministic) decapsulation algorithm outputs different shared keys on identical inputs.

$c = (c_1, c_2)$ such that $k \neq \bar{k}$ (with $k, \bar{k} \notin \{\perp\}$), where

$$\begin{aligned} k &\leftarrow \text{cKEM.DECAPS}(sk, c) \quad \text{and} \\ \bar{k} &\leftarrow \text{cKEM.DECAPS}(\bar{sk}, c). \end{aligned}$$

Note that the shared keys are computed as

$$\begin{aligned} k &\leftarrow \text{COMBINE}(k_1, k_2, F(pk_1, pk_2, c_1, c_2), \Xi) \quad \text{and} \\ \bar{k} &\leftarrow \text{COMBINE}(\bar{k}_1, \bar{k}_2, F(\bar{pk}_1, \bar{pk}_2, c_1, c_2), \Xi), \end{aligned}$$

where $f(\text{PK}, \cdot) = 0$ holds by assumption. Thus, to ensure different values for k and $\bar{k}$, we need that at least one of the following two inequalities holds: $k_2 \neq \bar{k}_2$ or $k_1 \neq \bar{k}_1$. Let $\mathbf{E}_1$, $\mathbf{E}_2$, and $\mathbf{E}_3$ denote the event that both inequalities, only the first inequality, and only the second inequality hold, respectively. Thus, we obtain

$$\begin{aligned} \text{Adv}_{\text{cKEM}[f]}^{\text{X-BIND-CT-K}}(\mathcal{A}) &= \Pr[\mathcal{A} \text{ wins X-BIND-CT-K}] \\ &= \sum_{i=1}^3 \Pr[\mathcal{A} \text{ wins X-BIND-CT-K} \wedge \mathbf{E}_i]. \end{aligned}$$

Let $\mathcal{B}_i, \mathcal{C}_i \in \{\mathcal{H}_i, \mathcal{L}_i\}$ be the generic reductions. We can now analyze the reductions w.r.t. the three aforementioned events.

Event $\mathbf{E}_1$ ($k_1 \neq \bar{k}_1 \wedge k_2 \neq \bar{k}_2$):

In this case, reduction $\mathcal{B}_i$ breaks X-BIND-CT-K security of KEM_i . This yields

$$\begin{aligned} &\Pr[\mathcal{A} \text{ wins X-BIND-CT-K} \wedge \mathbf{E}_1] \\ &\leq \min \left(\text{Adv}_{\text{KEM}_1}^{\text{X-BIND-CT-K}}(\mathcal{B}_1), \text{Adv}_{\text{KEM}_2}^{\text{X-BIND-CT-K}}(\mathcal{B}_2) \right). \end{aligned}$$

Event $\mathbf{E}_2$ ($k_1 = \bar{k}_1 \wedge k_2 \neq \bar{k}_2$):

In this case, reduction $\mathcal{C}_1$ breaks X-BIND-K,CT-PK security of KEM_1 while reduction $\mathcal{B}_2$ breaks X-BIND-CT-K security of KEM_2 . This yields

$$\begin{aligned} &\Pr[\mathcal{A} \text{ wins X-BIND-CT-K} \wedge \mathbf{E}_2] \\ &\leq \min \left(\text{Adv}_{\text{KEM}_1}^{\text{X-BIND-K,CT-PK}}(\mathcal{C}_1), \text{Adv}_{\text{KEM}_2}^{\text{X-BIND-CT-K}}(\mathcal{B}_2) \right). \end{aligned}$$

Event $\mathbf{E}_3$ ($k_1 \neq \bar{k}_1 \wedge k_2 = \bar{k}_2$):

In this case, reduction $\mathcal{B}_1$ breaks X-BIND-CT-K security of KEM_1 while reduction $\mathcal{C}_2$ breaks X-BIND-K,CT-PK security of KEM_2 . This yields

$$\begin{aligned} &\Pr[\mathcal{A} \text{ wins X-BIND-CT-K} \wedge \mathbf{E}_3] \\ &\leq \min \left(\text{Adv}_{\text{KEM}_1}^{\text{X-BIND-CT-K}}(\mathcal{B}_1), \text{Adv}_{\text{KEM}_2}^{\text{X-BIND-K,CT-PK}}(\mathcal{C}_2) \right). \end{aligned}$$

Collecting the above yields

$$\begin{aligned}
& \mathbf{Adv}_{\text{cKEM}[f]}^{\text{X-BIND-CT-K}}(\mathcal{A}) \\
&= \sum_{i=1}^3 \Pr[\mathcal{A} \text{ wins X-BIND-CT-K} \wedge \mathbf{E}_i] \\
&\leq \min \left(\mathbf{Adv}_{\text{KEM}_1}^{\text{X-BIND-CT-K}}(\mathcal{B}_1), \mathbf{Adv}_{\text{KEM}_2}^{\text{X-BIND-CT-K}}(\mathcal{B}_2) \right) \\
&\quad + \min \left(\mathbf{Adv}_{\text{KEM}_1}^{\text{X-BIND-K,CT-PK}}(\mathcal{C}_1), \mathbf{Adv}_{\text{KEM}_2}^{\text{X-BIND-CT-K}}(\mathcal{B}_2) \right) \\
&\quad + \min \left(\mathbf{Adv}_{\text{KEM}_1}^{\text{X-BIND-CT-K}}(\mathcal{B}_1), \mathbf{Adv}_{\text{KEM}_2}^{\text{X-BIND-K,CT-PK}}(\mathcal{C}_2) \right).
\end{aligned}$$

In more abstract form, our upper bound is of the form

$$\mathbf{Adv}_{\text{cKEM}[f]}^{\text{X-BIND-CT-K}}(\mathcal{A}) \leq \min(X_1, X_2) + \min(Y_1, Y_2) + \min(Z_1, Z_2), \quad (1)$$

where X_i , Y_i , and Z_i are placeholders for the specific advantages against the underlying KEM_i . For $i, j, k \in \{1, 2\}$, any sum $X_i + Y_j + Z_k$ is an upper bound for the X-BIND-CT-K advantage of $\mathcal{A}$ against cKEM. In particular, both $X_1 + Y_1 + Z_1$ and $X_2 + Y_2 + Z_2$ are upper bounds, hence also their minimum. In our specific case, we have $X_1 = Z_1$ and $X_2 = Y_2$, which yields the bound:

$$\begin{aligned}
& \mathbf{Adv}_{\text{cKEM}[f]}^{\text{X-BIND-CT-K}}(\mathcal{A}) \\
&\leq \min_{i \in \{1, 2\}} \left(2 \mathbf{Adv}_{\text{KEM}_i}^{\text{X-BIND-CT-K}}(\mathcal{B}_i) + \mathbf{Adv}_{\text{KEM}_i}^{\text{X-BIND-K,CT-PK}}(\mathcal{C}_i) \right) \quad (2)
\end{aligned}$$

Another upper bound for [Eq. \(1\)](#) is $\min(X_1, X_2) + Y_2 + Z_1$, which in our case yields

$$\begin{aligned}
\mathbf{Adv}_{\text{cKEM}}^{\text{X-BIND-CT-K}}(\mathcal{A}) &\leq \min_{i \in \{1, 2\}} \left(\mathbf{Adv}_{\text{KEM}_1}^{\text{X-BIND-CT-K}}(\mathcal{B}_1), \mathbf{Adv}_{\text{KEM}_2}^{\text{X-BIND-CT-K}}(\mathcal{B}_2) \right) \\
&\quad + \mathbf{Adv}_{\text{KEM}_1}^{\text{X-BIND-CT-K}}(\mathcal{B}_1) + \mathbf{Adv}_{\text{KEM}_2}^{\text{X-BIND-CT-K}}(\mathcal{B}_2) \\
&\leq 2 \mathbf{Adv}_{\text{KEM}_1}^{\text{X-BIND-CT-K}}(\mathcal{B}_1) + 2 \mathbf{Adv}_{\text{KEM}_2}^{\text{X-BIND-CT-K}}(\mathcal{B}_2), \quad (3)
\end{aligned}$$

where we replace the minimum by the sum of its arguments to simplify the bound.

Note that [Eq. \(2\)](#) is a bound in the “hybrid” sense, meaning we only need security of one underlying KEM (though both X-BIND-CT-K and X-BIND-K,CT-PK) to achieve X-BIND-CT-K security of the combined KEM. Alternatively, [Eq. \(3\)](#) shows that having X-BIND-CT-K for both underlying KEMs is also sufficient, thereby removing the requirement for X-BIND-K,CT-PK at the cost of requiring security for both KEMs.

Lastly, we consider the MAL setting, in which the adversary can choose the two key pairs $(pk, sk), (\bar{pk}, \bar{sk})$ with $pk = (pk_1, pk_2)$, $sk = (sk_1, sk_2)$, $\bar{pk} = (pk_1, \bar{pk}_2)$, and $\bar{sk} = (sk_1, sk_2)$ itself. To win, the outputs of $\mathcal{A}$ must result in *one* ciphertext $c = (c_1, c_2)$ for *two different* shared keys, i.e., $k \neq \bar{k}$ (with

$k, \bar{k} \notin \{\perp\}$). Then the same argumentation as given above for the HON and LEAK setting applies, with the following exceptions: In event E_2 and E_3 , the reductions do not necessarily break MAL-BIND-K,CT-PK security of KEM_1 and KEM_2 , respectively, as the adversary can choose these parts of the public keys to agree. Hence for $\mathcal{B}_i = \mathcal{M}_i$, the bounds from above simplify as follows

$$\begin{aligned}\Pr[A \text{ wins MAL-BIND-CT-K} \wedge E_2] &\leq \mathbf{Adv}_{KEM_2}^{\text{MAL-BIND-CT-K}}(\mathcal{B}_2), \\ \Pr[A \text{ wins MAL-BIND-CT-K} \wedge E_3] &\leq \mathbf{Adv}_{KEM_1}^{\text{MAL-BIND-CT-K}}(\mathcal{B}_1),\end{aligned}$$

which yields together with the E_1 -case, the final bound

$$\mathbf{Adv}_{\text{cKEM}[f]}^{\text{MAL-BIND-CT-K}}(\mathcal{A}) \leq 2 \mathbf{Adv}_{KEM_1}^{\text{MAL-BIND-CT-K}}(\mathcal{B}_1) + 2 \mathbf{Adv}_{KEM_2}^{\text{MAL-BIND-CT-K}}(\mathcal{B}_2).$$

This concludes the proof. $\square$

Remark 8. Note that the first bound given in [Theorem 7](#) for $X \in \{\text{HON}, \text{LEAK}\}$, i.e.,

$$\begin{aligned}\mathbf{Adv}_{\text{cKEM}[f]}^{X\text{-BIND-CT-K}}(\mathcal{A}) \\ \leq \min_{i \in \{1,2\}} \left(2 \mathbf{Adv}_{KEM_i}^{X\text{-BIND-CT-K}}(\mathcal{B}_i) + \mathbf{Adv}_{KEM_i}^{X\text{-BIND-K,CT-PK}}(\mathcal{C}_i) \right),\end{aligned}$$

also follows from [Corollary 6](#), i.e.,

$$\mathbf{Adv}_{\text{cKEM}[f]}^{X\text{-BIND-CT-K}}(\mathcal{A}) \leq \min_{i \in \{1,2\}} \left(\mathbf{Adv}_{KEM_i}^{X\text{-BIND-CT-PK}}(\mathcal{B}_i) \right).$$

This can be seen using the fact that

$$X\text{-BIND-CT-PK} \Leftrightarrow (X\text{-BIND-CT-K,PK} \wedge X\text{-BIND-K,CT-PK}),$$

which follows from [\[CDM24, Lemma 4.4 and Theorem 4.5\]](#) and the implication $X\text{-BIND-CT-K} \Rightarrow X\text{-BIND-CT-K,PK}$, which holds in the HON and LEAK setting by [Lemma 3](#).⁷

The following theorem shows that having LEAK-BIND-CT-K security for only one of the underlying KEMs is insufficient for the combined KEM to inherit this security notion.

Theorem 9. *Let KEM_1 and KEM_2 be two KEMs. Let further cKEM denote the KEM resulting from applying GC to KEM_1 and KEM_2 . Assume that KEM_1 is LEAK-BIND-CT-K secure but not LEAK-BIND-K,CT-PK secure, i.e., there exists an adversary $\mathcal{A}_1$ that breaks LEAK-BIND-K,CT-PK of KEM_1 with non-negligible probability δ_1 . Assume further that KEM_2 is not LEAK-BIND-CT-K secure, i.e., there exists an adversary $\mathcal{A}_2$ that breaks LEAK-BIND-CT-K security of KEM_2 with non-negligible advantage δ_2 . Then there exists an adversary $\mathcal{A}$ such that*

$$\mathbf{Adv}_{\text{cKEM}}^{\text{LEAK-BIND-CT-K}}(\mathcal{B}) \geq \delta_1 \delta_2 - \text{negl}.$$

⁷ Note that the notion X-BIND-CT-K,PK is not explicitly mentioned in the framework from [\[CDM24\]](#). However, it easily arises as the special case of the notion X-BIND-CT-PK, additionally requiring that the shared keys have to differ.

Proof. The idea of the proof is as follows. Adversary $\mathcal{A}$ —receiving two combiner key pairs, each consisting of one key pair per underlying KEM—simply runs the adversaries against the underlying KEMs. By assumption, both output a *single* ciphertext that decrypts validly under both key pairs; this condition is necessary to ensure that cKEM does not output $\perp$. Since one of the adversaries (in our case $\mathcal{A}_2$ which breaks LEAK-BIND-CT-K of KEM_2) outputs a ciphertext that decrypts to *different* shared keys, this is sufficient to ensure that COMBINE receives different inputs, thus resulting in different shared keys for the combined KEM.

Let $\mathcal{A}$ be an adversary against the LEAK-BIND-CT-K security of cKEM. It requests two honestly generated key pairs $(pk, sk), (\overline{pk}, \overline{sk})$, where $pk = (pk_1, pk_2)$, $sk = (sk_1, sk_2)$, $\overline{pk} = (\overline{pk}_1, \overline{pk}_2)$, and $\overline{sk} = (\overline{sk}_1, \overline{sk}_2)$, as input. With overwhelming probability, the individual key pairs are distinct, i.e., $(pk_1, sk_1) \neq (\overline{pk}_1, \overline{sk}_1)$ and $(pk_2, sk_2) \neq (\overline{pk}_2, \overline{sk}_2)$. Then, $\mathcal{A}$ runs $\mathcal{A}_1((pk_1, sk_1), (\overline{pk}_1, \overline{sk}_1))$ to obtain c_1 and $\mathcal{A}_2((pk_2, sk_2), (\overline{pk}_2, \overline{sk}_2))$ ⁸ to obtain c_2 . Finally, $\mathcal{A}$ outputs the ciphertext $c \leftarrow (c_1, c_2)$.

By assumption on the adversaries $\mathcal{A}_1$ (against LEAK-BIND-K,CT-PK) and $\mathcal{A}_2$ (against LEAK-BIND-CT-K), the following two equations hold with probability δ_1 and δ_2 , respectively:

$$\begin{aligned} \perp &\neq \text{KEM}_1.\text{DECAPS}(sk_1, c_1) = \text{KEM}_1.\text{DECAPS}(\overline{sk}_1, c_1) \neq \perp \\ \perp &\neq \text{KEM}_2.\text{DECAPS}(sk_2, c_2) \neq \text{KEM}_2.\text{DECAPS}(\overline{sk}_2, c_2) \neq \perp \end{aligned}$$

For $i \in \{1, 2\}$, let $k_i = \text{KEM}_i.\text{DECAPS}(sk_i, c_i)$ and $\overline{k}_i = \text{KEM}_i.\text{DECAPS}(\overline{sk}_i, c_i)$. By construction of cKEM, it holds that

$$k = \text{cKEM}.\text{DECAPS}(sk, c) = \text{COMBINE}(k_1, k_2, F(pk_1, pk_2, c_1, c_2), \Xi)$$

and

$$\overline{k} = \text{cKEM}.\text{DECAPS}(\overline{sk}, c) = \text{COMBINE}(\overline{k}_1, \overline{k}_2, F(\overline{pk}_1, \overline{pk}_2, c_1, c_2), \Xi).$$

Since $k_2 \neq \overline{k}_2$, we have $k \neq \overline{k}$ except the inputs form a collision for COMBINE—which happens only with negligible probability. Collecting everything then yields

$$\text{Adv}_{\text{cKEM}}^{\text{LEAK-BIND-CT-K}}(\mathcal{B}) \geq \delta_1 \delta_2 - \text{negl},$$

which concludes the proof. $\square$

Note that the above theorem holds regardless of the choice of f . Thus, it is also applicable in case of $f(\text{PK}, \cdot) = 0$ as was the case for [Theorem 7](#), thereby showing that having LEAK-BIND-CT-K security on only on underlying KEM is insufficient to achieve the same security on the combined KEM. Note further that [Theorem 9](#) requires a KEM that is LEAK-BIND-K,CT-PK secure but insecure w.r.t. LEAK-BIND-CT-K; Cremers et al. [[CDM24](#), Appendix B.6] show that KEMs with this property exist.

⁸ In theory, $\mathcal{A}$ can request to play against a *single* key pair. However, in this case, it effectively plays LEAK-BIND-CT,PK-K which is trivially achieved by *any* KEM. Hence, w.l.o.g. we can assume that $\mathcal{A}_2$ requests two different key pairs.

Table 1. Overview of combiners regarding their optional inputs to COMBINE—the shared keys k_1 and k_2 are always inputs. Note further that subscripts 1 and 2 refer to the classical and post-quantum KEM in the specifications, respectively.

Combiner	f	Optional inputs to COMBINE			
		pk_1	c_1	pk_2	c_2
CHEMPAT [Jos25]	$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$	✓	✓	✓	✓
GHP [CBG25]	$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$	✓	✓	✓	✓
QSF [BCD ⁺ 24]	$\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$	✓	✓	✗	✗
LAMPS [OGP ⁺ 25]	$\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$	✓	✓	✗	✗
OPENPGP [KRSW25]	$\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$	✓	✓	✗	✗
TLS [SFG25]	$\begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$	✗	✓	✗	✓
QSI [CHH ⁺ 25]	$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$	✗	✗	✗	✗

4 Real-World KEM-Combiners

In this section, we apply our results to real-world combiners. We describe the combiners under consideration in Section 4.1. In Section 4.2, we instantiate the combiners and give concrete results regarding the binding security of the resulting KEMs. In Section 4.3, we check what combiner properties are required when combining concrete KEMs.

4.1 Description of the KEM-Combiners

We consider a number of KEM combiners, which are currently described as IETF drafts. This includes the combiners GHP [CBG25], CHEMPAT [Jos25], and LAMPS [OGP⁺25], as well as the combiners used in the protocols OPENPGP [KRSW25] and TLS [SFG25]—we follow the convention that the combiner is simply named after the protocol in which it is used. Note that we consider the combiners instantiated with various KEMs, while some are originally only defined for a certain KEMs according to their specification, e.g., GHP [CBG25] uses ML-KEM for KEM₂.

X-WING is another scheme, for which there is an IETF draft [CSW25]. While X-WING comes instantiated with ML-KEM and DH-KEM (using the curve X25519), there is also a more generic version called QSF [BCD⁺24], which only fixes the elliptic curve component. Moreover, recently, the generic combiner QSI [CHH⁺25] was introduced, which works for any two KEMs—though additional requirements for the KEMs have to be met. An overview of the combiners under consideration is given in Table 1.

4.2 Binding Security of Real-World KEM-Combiners

In this section, we analyze combiner instantiations with respect to their binding security using our results from Section 3. While in the past, mostly classical and

post-quantum schemes were combined, we also consider instantiations using two post-quantum schemes. With their ongoing development and standardization, there is now a number of post-quantum schemes suited for combination: there are schemes that enjoy more trust due to being older (CLASSIC-McELIECE) or (soon to be) standardized (ML-KEM and HQC). We consider the post-quantum schemes ML-KEM, FRODOKEM, CLASSIC-McELIECE, BIKE, HQC, and HQC*—a modified variant of HQC proposed in [KSW25] that achieves more binding properties.⁹ This selection contains the schemes chosen by NIST for standardization, i.e., ML-KEM and HQC, as well as the finalists in NISTs standardization process, i.e., CLASSIC-McELIECE and BIKE.¹⁰ Regarding classical schemes, we include only DH-KEM as it is, to our knowledge, the only classical scheme for which the binding properties have been analyzed [CDM24]. More generally, the KEMs we consider are the ones for which complete binding analyses exist.

Note that all schemes under consideration use implicit rejection, which implies that the notions X-BIND-CT-PK and X-BIND-CT-K can never be achieved. Moreover, we focus on achieving the HON and LEAK variants of the notions as they seem the most relevant from a practical perspective—both known attacks exploit the absence of certain LEAK notions [CDM24, BJKS24]. This leaves us with the notions X-BIND-K-CT and X-BIND-K-PK for $X \in \{\text{HON}, \text{LEAK}\}$ as these, by the hierarchy, imply the remaining ones (X-BIND-K,PK-CT and X-BIND-K,CT-PK).

These notions are covered by Theorem 4, which can be applied to obtain the results for concrete combiner instantiations. The theorem states that a combined KEM is X-BIND-P-Q secure, if the combiner function COMBINE is collision-resistant and each underlying KEM for which the Q-component is not included in COMBINE (meaning the corresponding entry for f is 0) is required to fulfill the property X-BIND-P-Q itself.

Of the considered KEM-combiners, most instantiate COMBINE using standard hash functions. More precisely, [Jos25, OGP⁺25, KRSW25] all mention SHA as an instantiation. In contrast, TLS relies on a dual-PRF [SFG25, BBF⁺19]. First of all, the dual-PRF property does not imply collision resistance. As an example, consider the case where a dual-PRF F is modified to a function F^* s.t. $F^*(k_1, k_2)$ outputs a fixed value whenever $k_1 = k_2$, and otherwise behaves like F ; clearly, F^* is not collision-resistant but still works like a dual-PRF as an adversary needs to query the (unknown) key to trigger the special case.

Regarding the binding properties of the KEMs under consideration, we rely on the results from [GMP22, CDM24, KSW25], which cover the KEMs ML-KEM, FRODOKEM, CLASSIC-McELIECE, BIKE, HQC, HQC*, and DH-KEM. We provide an overview of the results relevant for our analysis in Table 2.

Then, Theorem 4 and the existing binding analysis for the selected KEMs enable an easy analysis for concrete combiners. For example, in the case of the

⁹ Note that the HQC team adopted the HQC* variant in the newest version of their specification (from August 22, 2025).

¹⁰ Note that SIKE was another finalist, but it was found to be insecure during the standardization process.

TLS combiner, neither of the public keys of the underlying KEMs is included in COMBINE, i.e., the combiner only achieves LEAK-BIND-K-PK if both KEMs do. In particular, this means that the TLS combiner applied to DH-KEM and ML-KEM would yield an LEAK-BIND-K-PK secure scheme (as both KEMs achieve LEAK-BIND-K-PK security), while this would not be the case for DH-KEM and HQC (as HQC does not fulfill LEAK-BIND-K-PK security). Below we describe our results for the combiners under consideration, which are shown in Table 1. We organize combiners with identical or similar outcomes into four groups.

CHEMPAT and GHP. Both combiners include shared keys, public keys, and ciphertexts of both underlying KEMs in COMBINE. Hence, Theorem 4 implies that only collision resistance of COMBINE is required to achieve LEAK-BIND-K-CT and LEAK-BIND-K-PK security for the combined KEM. In particular, binding security of CHEMPAT and GHP is guaranteed for *any* two KEMs.

QSF, LAMPS, and OPENPGP. These three combiners share a common structure, where for the first KEM *all* inputs are fed into COMBINE, while for the second one only the shared key is used. All three combiners require ciphertext second-preimage resistance (C2PRI) of the second KEM in order to achieve even IND-CCA security. Note that [BCD⁺24] shows that X-Wing requires ciphertext second-preimage resistance (C2PRI) of the second KEM in order to achieve even IND-CCA security. The C2PRI notion is very similar to LEAK-BIND-K,PK-CT with the only difference that the adversary has to find pre-images instead of collisions—in particular, LEAK-BIND-K,PK-CT implies C2PRI. Since all KEMs except HQC fulfill LEAK-BIND-K-CT and thus also LEAK-BIND-K,PK-CT, the assumption of C2PRI is given for all schemes except HQC.¹¹ Therefore, we cannot use HQC as an instantiation for the second KEM KEM₂. Regarding binding security, Theorem 4 yields that the second KEM has to achieve X-BIND-K-CT and X-BIND-K-PK security in order for the combined KEM to fulfill these notions. Taking this into account, Table 3 shows which KEM combinations are possible when using OPENPGP and LAMPS.

TLS. The draft [SFG25] describes a combiner that computes the shared key as $k = k_1 \parallel k_2$. In the subsequent TLS key schedule, further operations involving the ciphertexts are executed. Therefore, the TLS combiner can be seen to follow the dual-PRF approach from [BBF⁺19]—this is also explicitly stated in [SFG25]. Due to this, we consider the dual-PRF combiner, which uses the shared keys and ciphertexts of both KEMs as input to COMBINE. By Theorem 4, TLS requires X-BIND-K-PK security for both KEMs. We display the results in Table 5.

QSI. The QSI combiner includes only the shared keys of both underlying KEMs in COMBINE. Note that IND-CCA security of the combined KEM is only given

¹¹ While HQC is shown to be insecure w.r.t. LEAK-BIND-K,PK-CT, it could still fulfill the weaker notion C2PRI. However one easily checks that the LEAK-BIND-K,PK-CT attack from [KSW25] also applies against the notion C2PRI (as the shared key and one of the ciphertexts in the attack are honestly generated).

Table 2. Summary of the binding analysis of KEMs given in [GMP22, CDM24, KSW25]. The entries show if a binding notion holds in the LEAK and the HON setting (✓), only in the HON setting (✓), or in neither setting (✗).

KEM	X-BIND-K-CT	X-BIND-K-PK
ML-KEM	✓	✓
HQC	✗	✓
HQC*	✓	✓
DH-KEM	✓	✓
FRODOKEM	✓	✓
BIKE	✓	✓
McELIECE	✓	✗

if one of the underlying KEMs is IND-CCA secure, while the other one fulfills C2PRI. Due to the fact that only one of the KEMs has to fulfill C2PRI and only HQC does not fulfill it, we can consider all combinations of our selected KEMs.¹² Regarding binding security, Theorem 4 implies that both KEMs have to fulfill X-BIND-K-CT and X-BIND-K-PK. Hence, fewer combinations are possible for the QSI combiner—the details can be found in Table 4.

4.3 How to Combine Specific KEMs

In the previous section, we considered various combiners instantiated with the KEMs for which binding properties have been analyzed. The tables (cf. Table 3, Table 4, and Table 5) show, for each combiner, which combinations of KEMs result in a combined KEM achieving all binding notions.¹³ Here, we take a different perspective and consider which components of the two KEMs really need to be inputs to COMBINE. It turns out, that none of the considered combiners is optimal in the sense that it only feeds the minimum of components to COMBINE. Table 2 shows that all KEMs, except HQC, satisfy X-BIND-K-CT, which makes hashing the ciphertext unnecessary—at least from the perspective of achieving binding properties. On the other hand, several KEMs fall short regarding X-BIND-K-PK: HQC, HQC*, and BIKE achieve it only for $X = \text{HON}$, while McELIECE does not achieve it at all. This means that for several combinations, feeding the public key to COMBINE is more important than the ciphertext. However, none of the combiners provide solely the public key(s) to COMBINE. The OPENPGP combiner gives public key and ciphertext of one underlying KEM to COMBINE, while TLS gives the ciphertexts of both underlying KEMs. The results are summarized in Table 6 which shows, for each combination, what f needs to look like to achieve all binding properties.

¹² It is important to note that any combination of the form $X + \text{HQC}$ no longer achieves IND-CCA security if HQC turns out to not be IND-CCA—even if X ’s IND-CCA security remains intact.

¹³ Recall that we focus on security up to the LEAK setting and, since all KEMs are implicitly-rejecting, only on X-BIND-K-PK and X-BIND-K-CT.

Table 3. The binding results for KEMs obtained using one of the combiners QSF [BCD⁺24], OPENPGP [KRSW25], or LAMPS [OGP⁺25], where $F(pk_1, pk_2, c_1, c_2) = (pk_1, c_1)$. We consider instantiations using any combination of ML-KEM, HQC, HQC*, DH-KEM, FRODOKEM, BIKE, and CLASSIC-MCELIECE (shortened to MCELIECE in the table)—note that we exclude combinations of a scheme with itself (■). The entries show if security (w.r.t. both X-BIND-K-CT and X-BIND-K-PK) holds in the LEAK and the HON setting (✓), only in the HON setting (✓), or in neither setting (✗).

$\begin{array}{c} \text{KEM}_2 \\ \text{KEM}_1 \end{array}$	ML-KEM	HQC ^a	HQC*	DH-KEM	FRODOKEM	BIKE	MCELIECE
ML-KEM	■	✗	✓	✓	✓	✓	✗
HQC	✓	■	■	✓	✓	✓	✗
HQC*	✓	■	■	✓	✓	✓	✗
DH-KEM	✓	✗	✓	■	✓	✓	✗
FRODOKEM	✓	✗	✓	✓	■	✓	✗
BIKE	✓	✗	✓	✓	✓	■	✗
MCELIECE	✓	✗	✓	✓	✓	✓	■

^a The combiners require C2PRI security to achieve IND-CCA security. Since HQC fails to achieve it, any instantiation using $\text{KEM}_2 = \text{HQC}$ falls short not just w.r.t. binding security but security in general.

Table 4. The binding results for KEMs obtained using the combiner QSI [BCD⁺24], where $F(pk_1, pk_2, c_1, c_2) = \varepsilon$. We consider instantiations using any combination of ML-KEM, HQC, HQC*, DH-KEM, FRODOKEM, BIKE, and CLASSIC-McELIECE (shortened to McELIECE in the table)—note that we exclude combinations of a scheme with itself (■). The entries show if security (w.r.t. both X-BIND-K-CT and X-BIND-K-PK) holds in the LEAK and the HON setting (✓), only in the HON setting (✓), or in neither setting (✗).

$\begin{array}{c} \text{KEM}_2 \\ \backslash \\ \text{KEM}_1 \end{array}$	ML-KEM	HQC	HQC*	DH-KEM	FRODOKEM	BIKE	McELIECE
ML-KEM	■	✗	✓	✓	✓	✓	✗
HQC	✗	■	■	✗	✗	✗	✗
HQC*	✓	■	■	✓	✓	✓	✗
DH-KEM	✓	✗	✓	■	✓	✓	✗
FRODOKEM	✓	✗	✓	✓	■	✓	✗
BIKE	✓	✗	✓	✓	✓	■	✗
McELIECE	✗	✗	✗	✗	✗	✗	■

Table 5. The binding results for KEMs obtained using the TLS combiner [SFG25], where $F(pk_1, pk_2, c_1, c_2) = (c_1, c_2)$. We consider instantiations using any combination of ML-KEM, HQC, HQC*, DH-KEM, FRODOKEM, BIKE, and CLASSIC-McELIECE (shortened to McELIECE in the table)—note that we exclude combinations of a scheme with itself (■). The entries show if security (w.r.t. both X-BIND-K-CT and X-BIND-K-PK) holds in the LEAK and the HON setting (✓), only in the HON setting (✓), or in neither setting (✗).

$\begin{array}{c} \text{KEM}_2 \\ \backslash \\ \text{KEM}_1 \end{array}$	ML-KEM	HQC	HQC*	DH-KEM	FRODOKEM	BIKE	McELIECE
ML-KEM	■	✓	✓	✓	✓	✓	✗
HQC	✓	■	■	✓	✓	✓	✗
HQC*	✓	■	■	✓	✓	✓	✗
DH-KEM	✓	✓	✓	■	✓	✓	✗
FRODOKEM	✓	✓	✓	✓	■	✓	✗
BIKE	✓	✓	✓	✓	✓	■	✗
McELIECE	✗	✗	✗	✗	✗	✗	■

Table 6. Overview of which components need to be given as input to COMBINE to achieve all binding properties for any combination of two KEMs. Entries are in matrix form $\begin{bmatrix} pk_1 & pk_2 \\ c_1 & c_2 \end{bmatrix}$. A 0 means the component is not required to be an input to COMBINE, while a 1 means that the component has to be given as input to COMBINE. A $\star$ indicates that the component needs to be input to COMBINE to achieve LEAK security while otherwise HON security is achieved.

$\begin{matrix} \text{KEM}_2 \\ \text{KEM}_1 \end{matrix}$	ML-KEM	HQC	HQC*	DH-KEM	FRODOKEM	BIKE	McELIECE
ML-KEM	■	$\begin{bmatrix} 0 & \star \\ 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & \star \\ 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & \star \\ 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$
HQC	$\begin{bmatrix} \star & 0 \\ 1 & 0 \end{bmatrix}$	■	■	$\begin{bmatrix} \star & 0 \\ 1 & 0 \end{bmatrix}$	$\begin{bmatrix} \star & 0 \\ 1 & 0 \end{bmatrix}$	$\begin{bmatrix} \star & \star \\ 1 & 0 \end{bmatrix}$	$\begin{bmatrix} \star & 1 \\ 1 & 0 \end{bmatrix}$
HQC*	$\begin{bmatrix} \star & 0 \\ 0 & 0 \end{bmatrix}$	■	■	$\begin{bmatrix} \star & 0 \\ 0 & 0 \end{bmatrix}$	$\begin{bmatrix} \star & 0 \\ 0 & 0 \end{bmatrix}$	$\begin{bmatrix} \star & \star \\ 0 & 0 \end{bmatrix}$	$\begin{bmatrix} \star & 1 \\ 0 & 0 \end{bmatrix}$
DH-KEM	$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & \star \\ 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & \star \\ 0 & 0 \end{bmatrix}$	■	$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & \star \\ 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$
FRODOKEM	$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & \star \\ 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & \star \\ 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$	■	$\begin{bmatrix} 0 & \star \\ 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$
BIKE	$\begin{bmatrix} \star & 0 \\ 0 & 0 \end{bmatrix}$	$\begin{bmatrix} \star & \star \\ 0 & 1 \end{bmatrix}$	$\begin{bmatrix} \star & \star \\ 0 & 0 \end{bmatrix}$	$\begin{bmatrix} \star & 0 \\ 0 & 0 \end{bmatrix}$	$\begin{bmatrix} \star & 0 \\ 0 & 0 \end{bmatrix}$	■	$\begin{bmatrix} \star & 1 \\ 0 & 0 \end{bmatrix}$
McELIECE	$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 1 & \star \\ 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & \star \\ 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 1 & \star \\ 0 & 0 \end{bmatrix}$	■

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